WORKSHEET

Chapter 7: Integration of $\frac{1}{x}$

1 Find an expression for the following integrals.

a $\int \frac{1}{5x + 7} dx$

b $\int \frac{3}{6x - 5} dx$

c $\int \frac{12}{4x - 9} dx$

d $\int \frac{-6}{9x + 2} dx$

2 Find an expression for the following integrals.

a $\int \frac{4x + 3}{2x^2 + 3x} dx$

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b $\int \frac{4x^2 - 2}{2x^3 - 3x} dx$

c $\int \frac{3x - 2}{3x^2 - 4x + 2} dx$

d $\int \frac{2x^2 + x}{4x^3 + 3x^2} dx$

3 Find expressions for the following integrals.

a $\int \frac{3x}{3x^2 + 5} dx$

b $\int \frac{\sec^2(x)}{2\tan(x)} dx$

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c $\int \frac{4e^{2x}}{e^{2x}+4}dx$

d $\int \frac{12x^2+8}{3x^3+6x}dx$

4 Find an expression for $\int \frac{3+5x-6x^2-7x^3}{2x^2}dx$.

5 Show that $\int \left\{[\ln(x)]^2 \frac{1}{x}\right\}dx = \frac{1}{3}[\ln(x)]^3 + c$.

6 Show that $\int \left\{\sin[\ln(x)]\frac{1}{x}\right\}dx = -\cos[\ln(x)] + c$.

7 Show that $\int \tan(6x)dx = \frac{-1}{6}\ln[\cos(6x)] + c$.

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8 Consider the integral $\int \frac{1}{x\ln(x)}dx$.

Use the substitution $u = \ln(x)$ to show that $\int_2^4 \frac{1}{x\ln(x)}dx = \ln(2)$.

9 Show that $\int_1^e \frac{[\ln(x)]^2}{x}dx = \frac{1}{3}$.

10 Show that $\int_2^3 \frac{2x}{x^2 - 1}dx = \ln(8) - \ln(3)$

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Answers

1 **a** $\frac{1}{5}\ln(|5x+7|)$

b $\frac{1}{2}\ln(|6x-5|)$

c $3\ln(|4x-9|)$

d $\frac{-2}{3}\ln(|9x+2|)$

2 **a** $\ln(|2x^2+3x|)$

b $\frac{2}{3}\ln(|2x^3-3x|)$

c $\frac{1}{2}\ln(|3x^2-4x+2|)$

d $\frac{1}{6}\ln(|4x^3+3x^2|)$

3 **a** $\frac{1}{2}\ln(|3x^2+5|)$

b $\frac{1}{2}\ln(|\tan(x)|)$

c $2\ln(|e^{2x}+4|)$

d $\frac{4}{3}\ln(|3x^3+6x|)$

4 $\frac{-3}{2x}+\frac{5\ln|x|}{2}-3x-\frac{7x^2}{4}+c$

5–10 Demonstrations.

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